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Launcher Tipoff Errors 31 March 2008

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Introduction

It is common to imagine a rocket's travel up its launcher rail as perfectly constrained except for the one obvious degree of freedom. When the rocket reaches the end of the launcher, it magically changes from constrained to unconstrained motion with no hitches. This mind set is not supported by the facts. The purpose of this memo is to set the record straight.

The Right Way

While there are many ways to design a launcher rail and its interface with rockets, the right way was discovered in classical antiquity (early 1950's). The process followed is to describe the right way, provide the rationale supporting the "right" assertion, and discuss the wrong ways together with rationale for why they are wrong.

Suspend the rocket beneath the rail.
The major argument in favor is when the riding lugs and rail part company, gravity, even if the rail is nearly vertical, helps to move the twain apart. This minimizes the possibility of recontact with concomitant tipoff torque. Best of all, this design enables simultaneous lug release. By contrast, if the rocket rides above the rail, gravity impedes separation and increases the likelihood of recontact.

Use simultaneous release lug designs
Nearly all sounding rockets are constrained to fly along the launcher rail with two riding lugs, one forward and one aft. It is not commonly known that simultaneous releasing lug designs are possible. The sketch on the next page shows how it can be done: This was the standard operating approach in antiquity.

Basically, the two lug flanges have different widths with the forward flange being the wider one. The slot in the rail bottom plate has two different widths. When the front lug drops free the aft lug also drops free at the same time.

Consequences of not using the right way

The current custom is to ride the rocket above the launcher rail with a lug design that releases both lugs at the free end of the rail. While the rocket is constrained by only the aft lug, any perturbation torque, primarily weight and thrust misalignment, acts to induce a rotation about the aft lug. Since all lug designs are intentionally loose-fitting to minimize running friction, there is no effective yaw constraint during this time. The aft lug can be analyzed as though it were a ball joint.

Tipoff Torques

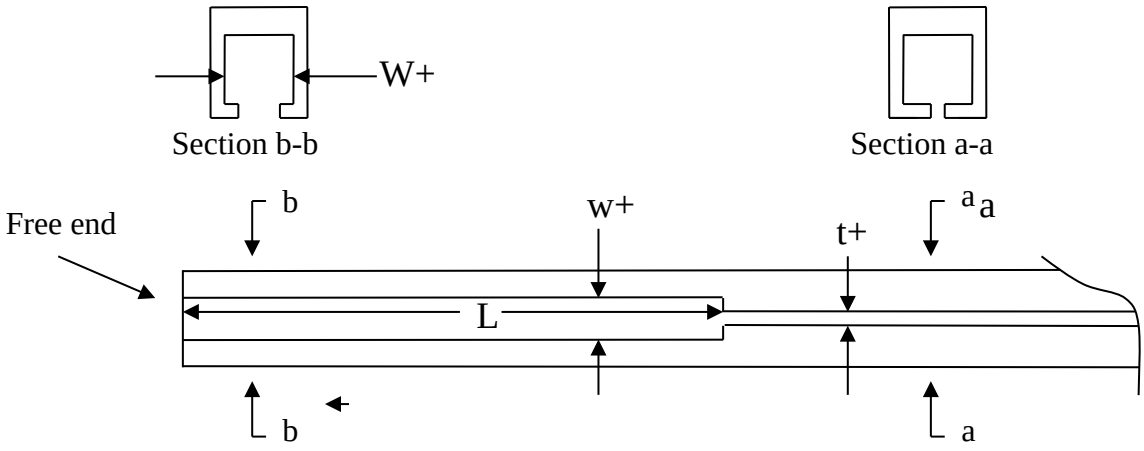
Weight

The weight torque is just

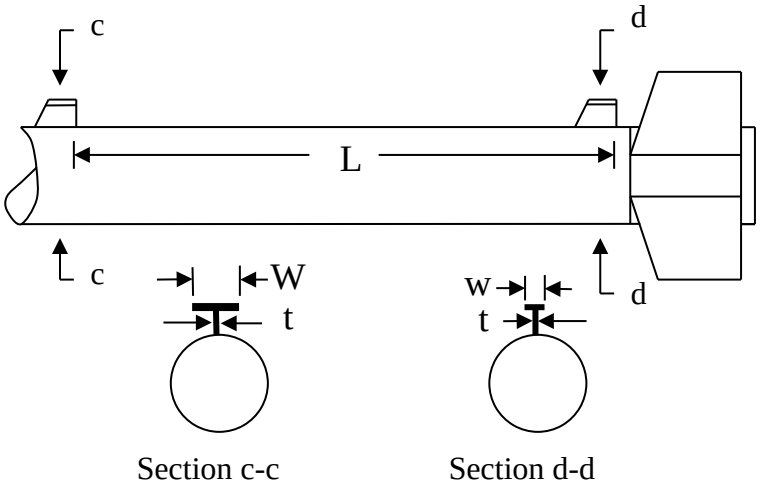
$$\tau = -W \cos(QE) * (x_{cg} - x_{l2}),$$

where τ = Torque,
W = Weight,
QE = Quadrant Elevation angle,
x_{cg} = Distance from the rocket nose tip to its CG, and
x_{l2} = Distance from the rocket nose tip to the center of its aft lug.

Note that there is no lateral or yaw component to this torque. It acts purely in pitch in a sense to reduce the QE to a value lower than the launch crew thought they were achieving. This torque is not in any way random; it happens the same way every time.



View of the bottom of the Rail looking up



Side view of Rocket and Riding Lugs

Thrust Misalignment

Thrust misalignment is a little trickier. First, because the longitudinal distance between nozzle throat and the aft lug is apt to be small, the **lδ** term in thrust misalignment torque can be neglected. But, there is no such relief for the cg offset torque. It is

$$\boldsymbol{\tau} = \mathbf{T}\,\boldsymbol{\Omega},$$

where **T** = Thrust, and
Ω = Lateral CG offset

The lateral CG offset is random with both the pitch and yaw components being normally distributed with zero mean.

Action time

Ordinary mechanics gives

$$\Delta t = \sqrt{2/a}\,(\sqrt{s_1 + L} + \sqrt{s_1}),$$

where Δt = Time during which only the aft lug acts to restrain motion,
a = Acceleration along the rail,
s_1 = Total distance traveled by the fron lug from first motion until release
L = Distance between the two lugs.

Tipoff Errors

The initial angular rate and angle of attack when the rocket has completely separated from its launcher rail are just

$$\omega = \tau \quad \Delta t / I_P, \text{ and}$$

$$\alpha = \theta = \tfrac{1}{2}\tau \quad \Delta t^2 / I_P,$$

where ω = Pitch/yaw rate,
α = Angle of attack/sideslip,
θ = Pitch/yaw angle, and
I_P = Pitch/yaw moment of inertia.